

Lab 12

EGRE246

This week you will get some experience working with the hash table.

1. In this lab you will implement a hash table with open hashing (separate chaining). The hash table will store positive integers. The hash function to be used is $h(x) = x \% M$ (the remainder after dividing x by M), where M is the size of the hash table. The set of possible keys x is the set of positive integers representable by the `int` data type.
2. You need to randomly generate more than 100 keys to insert your hash table and randomly search more than 10 keys from the hash table. You can use `rand` function (in `stdlib.h`) to generate random numbers.
3. Your hash table needs to implement delete function as well.
4. You can start from the code generated by ChatGPT, however, you need to make it right and fix the bugs from generated code.
5. Finishing up by coding and testing all the functions.
6. Run of completed project and submit your source code including `.h` and `.c` files on Canvas.